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Monsoon, 2025

B.Tech 5th Semester Examination

Environmental Engineering

Course Code: BCE23502T

Full Marks – 50

Time – 2 hours

The figure in the margin indicates full marks for the questions.

SECTION 1

Answer all the questions

(Multiple Choice) Marks: 10 x 1 = 10

- I. The term “potable water” refers to:
 - a) Water used for irrigation
 - b) Water used for industrial purpose
 - c) Water fit for human consumption
 - d) Water from rainfall
- II. Which of the following is a primary purpose of aeration in water treatment?
 - a) To remove suspended solids
 - b) To remove dissolved gases like CO₂ and H₂S
 - c) To remove microorganisms
 - d) To add chlorine to water
- III. The term ‘sullage’ refers to:
 - a) Industrial wastewater
 - b) Rainwater
 - c) Wastewater from kitchens and bathrooms
 - d) Sewage from toilets
- IV. The average rate of flow of domestic sewage is approximately:
 - a) 50% of water supply
 - b) 80% of water supply
 - c) Equal to water supply
 - d) Twice the water supply
- V. Which of the following represents a suspended growth treatment system?
 - a) Trickling filter
 - b) Activated sludge process
 - c) Biofilm reactor
 - d) Rotating biological contactor
- VI. Which type of pipe is most commonly used for internal water supply in buildings?
 - a) Cast iron pipe
 - b) PVC pipe
 - c) Concrete pipe
 - d) Stoneware pipe
- VII. The main constituents of municipal solid waste are:
 - a) Only plastics and metals
 - b) Organic matter, paper, plastics, glass, metals, etc.
 - c) Only biodegradable waste
 - d) Industrial hazardous waste

- VIII. The main difference between BOD and COD is that COD:
- Measures biological oxidation only
 - Takes longer to determine than BOD
 - Measures total oxygen required for chemical oxidation
 - Is always less than BOD
- IX. The A 1-L BOD bottle contains 20 mL of wastewater sample and the rest diluent. Initial DO in the bottle = 8.4 mg/L and DO after 5 days = 3.2 mg/L. What is the BOD₅ of the wastewater (mg/L)?
- 52
 - 260
 - 104
 - 26
- X. Which pollutant is primarily responsible for photochemical smog and is formed by reactions between NO_x and volatile organic compounds (VOCs) in sunlight?
- Sulfur dioxide (SO₂)
 - Ozone (O₃)
 - Carbon monoxide (CO)
 - Particulate matter (PM₁₀)

SECTION 2

1. Answer any one question from this section

Marks: 10

A) Explain the types of system of drainage used for carrying sewage. Also suggest a suitable system of drainage for an arid region in India.

B) What do you understand by filtration? Also discuss the action of filter in the process filtration.

SECTION 3

2. Answer any one question from this section

Marks: 10

A) Evaluate the effects of prolonged exposure to noise and air pollutants on human health and ecosystems.

B) Discuss the significance of BOD and COD in assessing sewage strength. Also, calculate the expected quantity of sewage for a population of 20,000 with an average sewage flow of 140 LPCD.

SECTION 4

3. Answer any one question from this section

Marks: 10

A) Design a slow sand filter for a water supply scheme to serve a population of 100000. The per capita demand is 150 LPCD and the rate of filtration is 250 L/hr/m².

B) A combined circular sewer is to be designed in a sewerage system for a city with a population of 1 lakh in an area of 100 hectares. The mean flow of sewage from the city is 250 LPCD. The rainfall intensity in that area is 4 cm/hr with an impermeability coefficient of 0.48. The ratio of peak to average sewage flow is 2.0. The Manning's roughness coefficient is 0.012. Determine the gradient of the sewer to carry the peak flow with a velocity of 1.2 m/s.

SECTION 5

4. Answer any one question from this section

Marks: 10

A) Explain the working mechanism and the types of sedimentations occurred in a sedimentation tank. Also design a sedimentation tank to treat 3 MLD water with a detention period of 4 hours and a velocity of flow 10 cm/min.

B) Compare suspended growth systems and attached growth systems with proper diagram.
